

Subspace Structure of WeatherGenerator Latent Tokens

Clustering methods and findings

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Abstract. We analyse the 2048-dimensional tokens the WeatherGenerator encoder produces for nine years of ERA5 states (160 M tokens). A K-subspaces clustering, a variant of k-means in which each cluster is a 64-dimensional affine subspace and tokens are assigned by their distance to that subspace, explains 68.5% of token variance with $K=128$ clusters. Plain k-means leaves 83.7% of the variance unexplained and produces a different partition. The clusters turn out to be climate regimes. They are geographically coherent, a minority of them carries a strong seasonal cycle, and the El Niño winters stand out in the central Pacific. Most importantly, the largest residual outliers of the fit coincide with documented extreme weather events, so the encoder does capture extremes.

1 Introduction

The WeatherGenerator (WGen) model encodes each 6-hourly ERA5 atmospheric state into 12,288 tokens, one per HEALPix grid cell, with 2048 dimensions each. This project analyses the encoder output for all 13,021 available states, covering 2014-01-01 00:00 to 2022-11-30 00:00 UTC. Two questions drove the analysis, in order of importance. First, can we tell from the latents whether the model actually encodes extreme weather events, meaning that real extremes show up as outliers? Second, how does the model handle spatial and temporal structure? Clustering answers both at once. It yields a regime label and a residual per cell and timestep, and the residual is the outlier score.

2 Methodology

Data. 13,021 encoder states of shape $[12,288 \times 2048]$, 1.2 TB in total. All clustering runs use the same fixed random sample of 7,000 states times all cells, 86 M tokens. Evaluation replays the fitted model on the full dataset or on held-out states.

K-subspaces. Cluster j is not a centre μ_j but an affine subspace (μ_j, U_j) with orthonormal basis $U_j \in \mathbb{R}^{2048 \times 64}$. The difference to k-means lies in the assignment step. A token x is assigned by its distance to the whole subspace,

$$r_j(x) = \underbrace{\|x - \mu_j\|^2}_{\text{k-means distance}} - \underbrace{\|U_j^\top (x - \mu_j)\|^2}_{\text{part the subspace explains}}, \quad (1)$$

the centre distance minus what the cluster’s own principal directions account for. A token far from μ_j along directions in which the cluster normally varies still belongs to it. The update step is exact PCA per cluster.

Algorithm (K-subspaces, $K=128$, $d=64$, $D=2048$)

init: $\mu_j \leftarrow K$ random tokens, $U_j \leftarrow 0$

repeat until $<0.1\%$ of labels change:

1. assign each token to the nearest subspace:

for every token x and cluster j :

$z \leftarrow U_j^\top (x - \mu_j)$ coords inside the subspace

$r_j(x) \leftarrow \|x - \mu_j\|^2 - \|z\|^2$ Eq. 1

label(x) $\leftarrow \arg \min_j r_j(x)$

2. refit each cluster to its members:

$\mu_j \leftarrow$ mean of its tokens

$U_j \leftarrow$ top- d eigenvectors (PCA) of their covariance

3. split the largest cluster to replace any collapsed one

Setting $d = 0$ turns the same code into plain k-means, the first baseline. Greedy k-center, which minimises the largest distance to a centre, is the second. A soft-assignment variant (an MPPCA mixture) was fitted afterwards and is used only to quantify per-token uncertainty. One fit takes about 2 hours on two A40 GPUs. Results were checked on held-out states, across three random restarts, and against an independent run at $K=256$ merged back to 128.

3 Results

3.1 Fit quality and baselines

At $K=128$, $d=64$ the fit splits token variance into 8.4% between cluster centres, 60.1% captured inside the subspaces, and 31.5% residual. So almost all of the structure is in each cluster’s directions of variation rather than in centre separation. On held-out dates the residual fraction is also 31.5%, meaning no overfitting.

Table 1 compares the methods. K-means leaves 83.7% of the variance unexplained. Its partition also differs from the subspace one (NMI **TBD**). Even when its clusters are given the same 64-dimensional PCA fit after the fact, the residual is still **TBD** per token against 1877 to 1888 for the subspace runs. Re-assigning tokens under the fitted subspace model by centre distance alone changes 57% of the labels and adds 20% residual. K-center is not competitive on this objective. Its greedy selection chases outliers, which yields cluster sizes from **TBD** and a minimax radius of **TBD**.

Table 1: Methods on the identical 86M-token sample, $K=128$. **obj/token** is the mean squared distance from a token to its assigned centre (subspace rows: to its assigned subspace). **% of var** is obj/token divided by the total token variance of 5999, the share of variance left unexplained, lower is better. **spread** is best minus worst obj/token across 3 random inits.

method	obj/token	% of var	spread
k-center (greedy)	TBD	TBD	
k-means ($d=0$)	5023.8	83.7%	TBD
K-subspaces ($d=64$)	1887.8	31.5%	0.24%
+ overcluster-merge	1877.0	31.3%	

3.2 Stability and the choice of K

Three restarts from different seeds land within 0.24% in objective and agree pairwise at NMI 0.68 to 0.72 (chance level 0.13). Fitting at $K=256$ and merging back to 128 with a Ward-style criterion improves the objective by 0.57%, which is 2.4 times the seed spread, while agreeing with the direct fit only at the seed-to-seed level. The merge cost curve has no knee anywhere between $K=256$ and 32, and 33% of tokens lie within a 10% residual margin of a second cluster. There is no natural number of regimes in this data. K is a resolution choice.

3.3 Extreme events are visible as outliers

The fit provides a residual per cell and timestep. Raw residuals are not comparable across cells, their per-cell mean spans a factor of 15.7, so each cell is standardised against its own history. The top 0.1% of standardised residuals are not noise. 89% of flagged cells have a flagged neighbour (3% for random cells) and 32% persist to the next 6 h step (0.01% random). The largest connected events match documented extremes: the February 2018 sudden-stratospheric-warming Arctic heat anomaly ($z=22.4$), the December 2015 above-freezing North Pole event ($z=19.5$), the August 2016 Arctic cyclone, and the March 2022 East-Antarctic heatwave. The March 2022 event, the largest temperature anomaly on record, sits below the global median in raw residual because the Antarctic plateau barely moves in latent space. Raw latent residual is not calibrated to physical anomaly size, only the standardised score recovers such events.

3.4 Locality: the clusters have territories

The clustering receives no location or time input, yet the dominant cluster per cell forms contiguous regions that follow tropics, storm tracks, polar caps and continents (Fig. 1). At raw presence level most clusters look global, a typical mid-size cluster appears at least once in a third of all cells, yet half of its tokens sit in only 4.4% of the globe. Across the run 77 of 128 clusters reach occupancy above 0.9 somewhere while 12 never pass 0.5. So clusters have clear territories that only a threshold reveals, which the newer per-cluster reports now apply. A few clusters encode something specific on their own. Cluster 27 owns the ring around the North

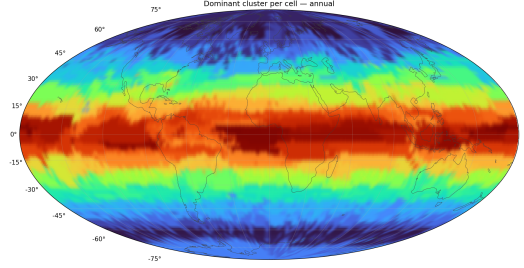


Figure 1: Dominant cluster per cell over the full year ($K=128$, similar clusters share hues), computed without any location input.

Pole between 75 and 86°N, the zone where sea ice is declining fastest, with occupancy 1.0 in its core cells. Its residual is nearly as large as the 6-hour change itself (ratio 0.92, third highest of 128), so the encoder gives this region its own regime but models it comparatively poorly.

3.5 Seasonality and El Niño

Seasonality is real but concentrated in a minority of clusters. The median seasonality score (std/mean of the twelve monthly enrichments) is 0.17, and only 35 of 128 clusters exceed 0.5, mostly winter or summer regimes. Whether this concentration is a property of the atmosphere or partly of the clustering method is open. At the level of the map, 42.8% of cells change their dominant cluster between DJF and JJA, while the diurnal counterpart is a null (2.6% between 00Z and 12Z). The leading principal component of the per-date regime mix is an almost pure seasonal harmonic ($R^2=0.992$). On top of the seasonal cycle, interannual events show: in the Niño 3.4 box the three El Niño winters of the period (2015/16, 2018/19, 2019/20) have a mean anomaly score of 0.55 to 0.60 while four of the five remaining winters sit at 0.12 to 0.18, and during the record 2015/16 event the cluster that usually dominates the box drops to its nine-year occupancy minimum.

4 Discussion

Both guiding questions have positive answers. Documented extremes (SSW 2018, North Pole melt 2015, Antarctic heatwave 2022) are recoverable from the latents as spatially and temporally coherent outlier events, with the caveat that raw residuals must be standardised per cell first because latent residual size is not calibrated to physical anomaly size. Spatially the latents are organised into regimes with real territories, which only become visible under a mass threshold since almost every cluster is present somewhere on the globe. Temporally the seasonal cycle dominates and is carried by a minority of clearly seasonal clusters, with El Niño winters visible on top of it. Occasionally a single cluster isolates a specific physical region, such as the Arctic sea-ice ring. Finally, the latent space is a continuum with no natural K , and its structure lies in per-cluster directions of variation, which is what required the subspace distance.